



Course Outcomes (CO):

CO1: To apply the knowledge of lex tool & yacc tool to develop a scanner & parser.

Assignment # 1_A

SECTION A (2*5=10)

1. Find the tokens for the given code:
`printf("i = %d, &i = %x", i, &i); l`
2. What is input buffering?
3. What is backpatching?
4. What is bootstrapping?
5. What is the difference between single pass compiler and multi pass compiler?

SECTION B (7*3=21 Mark)

1. Why do we need syntax tree while constructing compiler.
2. Explain Lexical Analyzer generator (LEX) and its structure.
3. What do you understand by the term cousins of compiler?

SECTION C (11*3=33 Mark)

1. Discuss compilation process. Write down the output of each phase for the expression
 $a = b * c + d/e$
2.
 - a. What is an ambiguous grammar? Specify the demerits of ambiguous grammar.
 - b. What are the various types of errors in compilation process? Explain each type with example.
3. What is Error handling? Discuss in detail.